

Quarto KEY

Justin Pomeranz

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Problems

Section 1: preliminaries

My level 2 header

My level 3 header

Section 2: Lists

2.1

My favorite animals

1. Bugs (obviously)
2. Here's another bug
3. And a third

2.2

Why Use Quarto

- You can make lots of outputs
- You can incorporate code/analysis/results directly in (don't need to copy/paste plots and stuff every time analysis changes)
- Focus on writing, not formatting

2.3

Sub Bulleted list

- Mountain Running
 - Get to see all the sites on a backpacking trip without having to carry so much stuff
 - Makes me feel cool
- Playing the banjo
 - Also makes me feel cool

Section 3: Equations

3.1

inline: $x \sim \text{Normal}(\mu = 0, \sigma = 1)$

3.2 Block equations:

$$Y = \beta_0 + \beta_1 * X +$$

3.3

$$HA : \mu_{reference} \neq \mu_{treatment}$$

3.4

$$s^2 = \frac{\sum (y_i - \bar{y})^2}{n - 1}$$

Section 4: Code Chunks

4.1

1. Add a code chunk which loads the tidyverse and reads in the surveys.csv file as an object called `finch` . At the end of the code chunk, display the `finch` object as a tibble

```
library(tidyverse)
surveys <- read.csv(here::here("data/surveys.csv"))
as_tibble(surveys)
```

A tibble: 35,549 x 9

	record_id	month	day	year	plot_id	species_id	sex	hindfoot_length	weight
	<int>	<int>	<int>	<int>	<int>	<chr>	<chr>	<int>	<int>
1	1	7	16	1977	2	NL	M	32	NA
2	2	7	16	1977	3	NL	M	33	NA
3	3	7	16	1977	2	DM	F	37	NA
4	4	7	16	1977	7	DM	M	36	NA
5	5	7	16	1977	3	DM	M	35	NA
6	6	7	16	1977	1	PF	M	14	NA
7	7	7	16	1977	2	PE	F	NA	NA

```

      8      8      7      16 1977      1 DM      M      37      NA
      9      9      7      16 1977      1 DM      F      34      NA
     10     10      7      16 1977      6 PF      F      20      NA
# i 35,539 more rows

```

4.2

2. Add a code chunk which uses the `group_by()` and `summarize()` functions to get a count of the number of individuals in each species ID, and display it as a tibble

```

surveys |>
  group_by(species_id) |>
  summarize(count = n())

```

```

# A tibble: 49 x 2
  species_id count
  <chr>      <int>
1 ""          763
2 "AB"         303
3 "AH"         437
4 "AS"           2
5 "BA"          46
6 "CB"          50
7 "CM"          13
8 "CQ"          16
9 "CS"           1
10 "CT"           1
# i 39 more rows

```

4.3

3. Add a code chunk which uses the `group_by()` and `summarize()` functions to get a count of the number of individuals in each species ID for each year, and display it as a tibble

```

surveys |>
  group_by(species_id, year) |>
  summarize(count = n())

```

```
# A tibble: 535 x 3
# Groups:   species_id [49]
  species_id year count
  <chr>      <int> <int>
1 ""        1977    16
2 ""        1978    56
3 ""        1979    61
4 ""        1980    40
5 ""        1981    55
6 ""        1982    14
7 ""        1983    21
8 ""        1984    30
9 ""        1985    22
10 ""       1986    20
# i 525 more rows
```

4.4

4. Add a code chunk which uses the `group_by()`, `filter()` and `summarize()` functions to get a mean weight for the D0 and the DM species ID and display it as a tibble

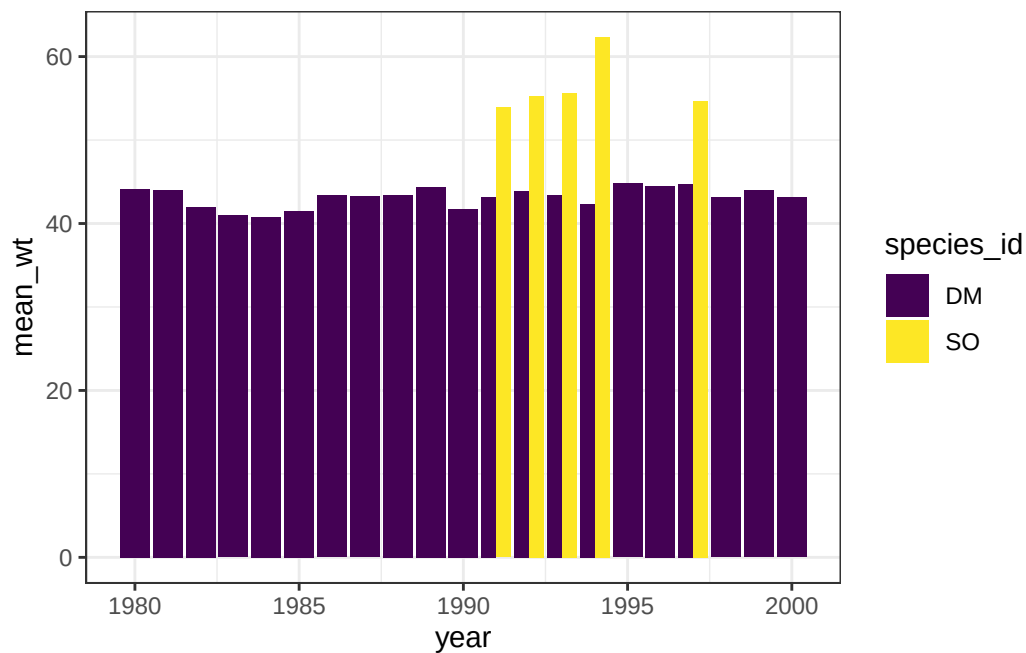
```
surveys |>
  group_by(species_id) |>
  filter(species_id == "D0" |
         species_id == "DM") |>
  summarize(mean_wt = mean(weight, na.rm = TRUE))
```

```
# A tibble: 2 x 2
  species_id mean_wt
  <chr>      <dbl>
1 DM        43.2
2 D0        48.9
```

4.5

5. Make a bar graph showing the mean weight for the S0 and the DM species for only the years 1980 through 2000. Make it so the fill of the bars changes color based on the two species IDs, the bar positions are “dodged” and not “stacked”. Finally, use a colorblind friendly palette and choose a theme for the plot that is not the default.

```
surveys |>
  group_by(species_id, year) |>
  filter(species_id == "SO" |
         species_id == "DM",
         year >= 1980,
         year <= 2000) |>
  summarize(mean_wt = mean(weight, na.rm = TRUE)) |>
  ggplot(aes(x = year,
             y = mean_wt,
             fill = species_id)) +
  geom_col(position = "dodge") +
  scale_fill_viridis_d() +
  theme_bw()
```



Problem 5: Miscellany

- 1.
2. my cmu email address is jpomeranz@colroadomesa.edu
3. Include a line of text which includes at least one

1. **bold**

2. *italicized*
3. subscript H₂O
4. superscript M²